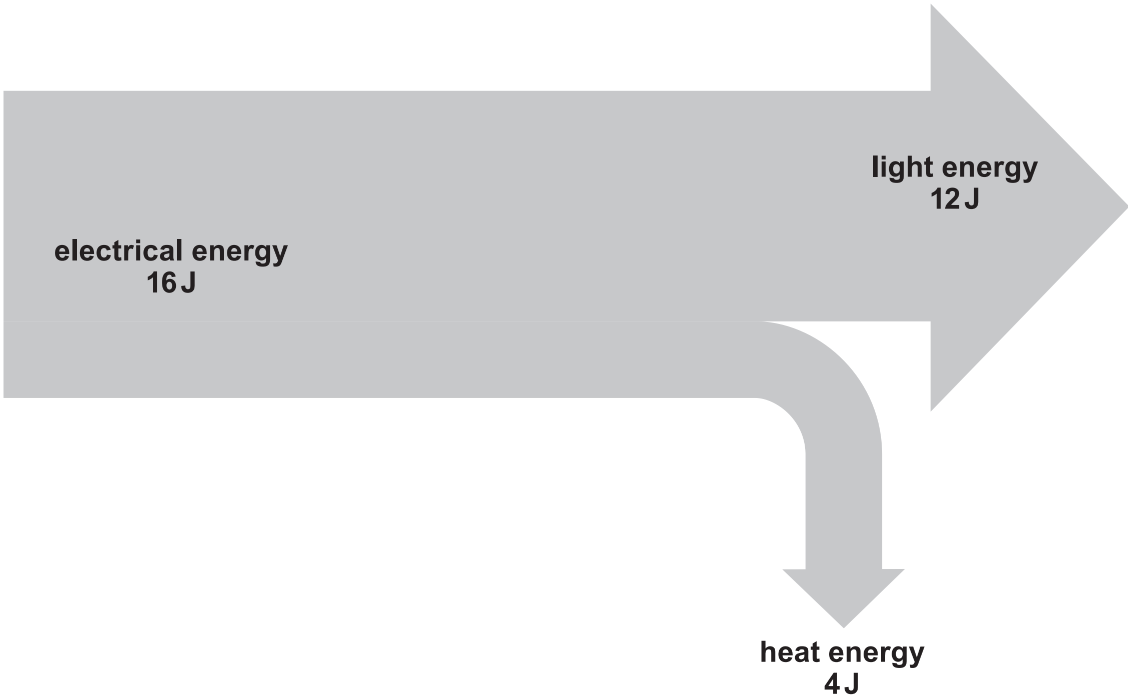
Examiner only

5. Lighting a home accounts for 18 % of a typical house's electricity bill.

(a) The diagram below shows energy flow through a LED lamp.



Use the equation:

$$\%\ \text{Efficiency} = \frac{\text{energy usefully transferred}}{\text{total energy}} \times 100$$

to calculate the efficiency of the LED lamp. [2]

% Efficiency = ...

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(b) The Energy Trust is an organisation in the UK that encourages householders to become more efficient in their energy use and to help to reduce their energy costs. The table below compares traditional filament and LED lamps.

	LED	Traditional filament lamp
cost of 1 lamp	£2.50	£1
lamp lifespan (hours)	50 000	1000
number of lamps used in 50 000 hours	1	....
cost of lamps used in 50 000 hours	£2.50	£ ....
power per lamp (kW)	0.01	0.06
units used in 50 000 hours (units = power × 50 000)	....	3 000
cost of one unit	20p	20p
cost of electricity for light for 50 000 hours = units used × cost of unit	£	£600
total cost of light for 50 000 hours	£102.50	£ ....

(i) **Complete** the table. [5]
(space for working)

(ii) The homeowner refuses to buy LED lamps because they are more expensive and he thinks he will lose money compared to using traditional filament lamps.

Explain whether you agree with the homeowner. [2]

....

....

....

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